

TEXT EDITOR

This project is a menu-driven Text Editor developed in C using the Doubly Linked List data structure.

It allows users to perform basic text editing operations such as insert, delete, search, replace, copy, paste, undo, and redo. The project also supports cursor navigation and file handling operations

like opening and saving text files. It is designed to demonstrate the practical implementation of linked lists,

stacks, and file handling concepts in Data Structures.

1. Insert Line

The Insert Line option is used to add a new line of text to the editor.

When the user enters a line, a new node is created using the doubly linked list, and the node is inserted at the end of the list. The inserted operation is also stored in the Undo stack so that it can be reversed later if required.

2. Delete Line

The Delete Line option removes a specified line from the editor.

The program searches for the required line, removes the corresponding node from the doubly linked list, and updates the previous and next links of the neighboring nodes.

The deleted information is stored in the Undo stack so that the deleted line can be restored using the Undo operation.

3. Display Text

The Display Text option displays all the lines currently present in the text editor.

It traverses the doubly linked list from the head node to the tail node and prints each line in the order in which it is stored.

4. Search Text

The Search Text option allows the user to search for a particular word or line in the editor.

The linked list is traversed sequentially, and each line is compared with the user input.

If the text is found, the program displays a success message; otherwise, it informs the user that the text is not present.

5. Replace Text

The Replace Text option is used to replace an existing word with a new word.

The program searches each line for the specified word and, if found, replaces it with the new word entered by the user.

After replacement, the updated line is stored back into the linked list.

6. Copy Line

The Copy Line option copies a selected line into a temporary clipboard without removing it from the editor.

The copied text remains available until another copy or cut operation is performed.

This clipboard is later used by the Paste operation.

7. Paste Line

The Paste Line option inserts the contents of the clipboard into the editor.

A new node is created with the copied or cut text and inserted into the linked list.

If the clipboard is empty, the program displays an appropriate message.

8. Undo

The Undo option reverses the most recent editing operation performed by the user.

It retrieves the latest action from the Undo stack, performs the opposite operation, and stores that action in the Redo stack. This allows the user to recover the undone action later if required.

9. Redo

The Redo option restores the operation that was previously undone.

It retrieves the latest action from the Redo stack, performs that action again, and stores it back in the Undo stack. This enables the user to reapply changes that were removed using Undo.

10. Move Cursor Up

The Move Cursor Up option moves the cursor to the previous line in the editor.

If the cursor is already at the first line, the program informs the user that no further upward movement is possible.

11. Move Cursor Down

The Move Cursor Down option moves the cursor to the next line in the editor.

If the cursor is already positioned at the last line, the program displays a message indicating that it cannot move further down.

12. Show Cursor

The Show Cursor option displays the current position of the cursor along with the text present at that position.

This helps the user identify the line on which the cursor is currently placed.

13. Open File

The Open File option loads the contents of an existing text file into the editor.

The program reads the file line by line, creates a node for each line, and stores all the lines in the doubly linked list so that they can be edited.

14. Save File

The Save File option stores the current contents of the editor into a text file.

The linked list is traversed from the beginning to the end, and every line is written to the specified file. This ensures that all modifications made by the user are permanently saved.

15. Close File

The Close File option closes the currently opened file after all necessary operations are completed.

It indicates that file-related operations have been finished successfully.

16. Exit

The Exit option terminates the execution of the text editor.

Before exiting, the user is expected to save any important changes to avoid losing data.

The program then closes all active operations and ends normally.